

Reproduction of attractivechaos/plb2: four CPU benchmarks in seven languages

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Independent reproduction · Benchmark or leaderboard entry · Programming languages and compilers

Abstract

Ran the plb2 programming-language benchmark at the pinned commit: the nqueen, matmul, sudoku and bedcov tasks in C (gcc 13, -O3 -flto), Rust, Go, Java, JavaScript (Node.js), Python (CPython 3.12) and Julia, compiled with each language's Makefile where one exists, one timed run per program on an Intel Xeon W-2145. Each program's last output line is kept in the log. At this commit the JavaScript sudoku exits with an error, and the Rust bedcov reports a different overlap count from the other languages.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	plb2, 7 languages x 4 tasks	2026-10-08	1	1	Signed by the K-Veritas server

2 Results

r1: plb2, 7 languages x 4 tasks

Run	Metric	Final value	Step
1	c_nqueen_seconds	2.8733	
1	c_matmul_seconds	1.8112	
1	c_sudoku_seconds	2.104	
1	c_bedcov_seconds	1.1856	
1	rust_nqueen_seconds	3.0356	
1	rust_matmul_seconds	1.7426	
1	rust_sudoku_seconds	2.2547	
1	rust_bedcov_seconds	1.3404	
1	go_nqueen_seconds	3.2979	
1	go_matmul_seconds	2.2567	
1	go_sudoku_seconds	2.4864	
1	go_bedcov_seconds	1.5523	

Run	Metric	Final value	Step
1	java_nqueen_seconds	4.3004	
1	java_matmul_seconds	1.8862	
1	java_sudoku_seconds	3.225	
1	java_bedcov_seconds	4.5043	
1	javascript_nqueen_seconds	6.575	
1	javascript_matmul_seconds	6.626	
1	javascript_bedcov_seconds	4.9239	
1	python_nqueen_seconds	132.682	
1	python_matmul_seconds	146.742	
1	python_sudoku_seconds	64.173	
1	python_bedcov_seconds	69.4029	
1	julia_nqueen_seconds	3.1641	
1	julia_matmul_seconds	1.8062	
1	julia_sudoku_seconds	2.8236	
1	julia_bedcov_seconds	2.2923	

Run	Inline claim	Value	Phase
1	c_nqueen_seconds	2.8733	
1	c_matmul_seconds	1.8112	
1	c_sudoku_seconds	2.104	
1	c_bedcov_seconds	1.1856	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	481.8 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	8969c6e5874897d1a2b4ad478f169c27421e800c4f1b0f272541adb20dbe733c	3.3 MB
r1.kvbundle.zip	29785ffc1967bbff75368401180ea7bc04ad8790a92a3e29cdf6f13e921a9ec5	60 KB

5 Links and licence

Reproduction of: <https://github.com/attractivechaos/plb2/tree/1181b3b83fea52ee85a66548845f8bf92e33fb67>

Record: CC-BY 4.0. Code: CC0-1.0, as declared by the authors. Tags: programming-languages, benchmark, performance.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.
- [2] Sealed experiment report r1 (plb2, 7 languages x 4 tasks), kv:2610.00024v1/r1. Data hash 55b93fb1f5dcea0f3f0e504fc005c1d06d93a4637461dc1f16b38f537f7bcf62.
<https://kveritas.org/records/2610.00024v1/r1.pdf>